ESD Main — motorized-fader MIDI controller
Rev 11 / generated from kicad/tools/esd_main_design.py
Authors: Nathan Cohen and Travis Nagle

Power

File: power.kicad_sch

MCU + bus

File: mcu_bus.kicad_sch

Memory-mapped I/O

File: mmio.kicad_sch

LCD

File: lcd.kicad_sch

MIDI + serial

File: midi_serial.kicad_sch

Faders + motor drivers

File: faders.kicad_sch

I2C + daughter bus

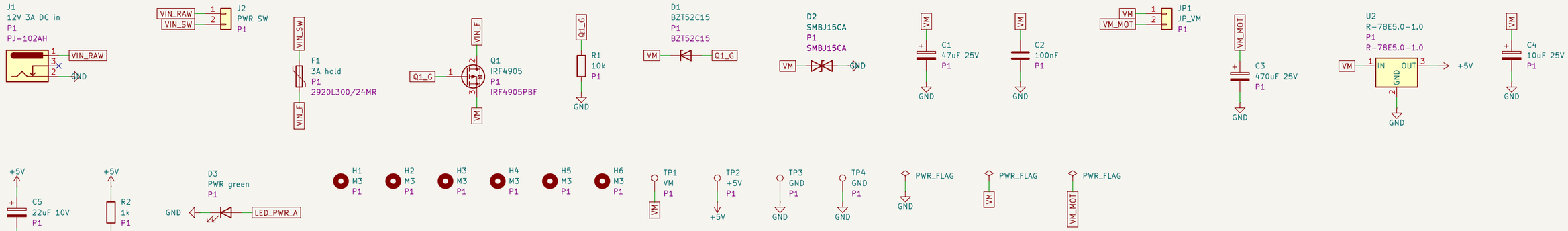
File: i2c_bus.kicad_sch

Authors: Nathan Cohen, Travis Nagle
Rev 11 / generated from kicad/tools/esd_main_design.py
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Sheet: /
File: ESD_Main.kicad_sch

Title: ESD Main

Size: A4	Date: 2026-10-19	**Rev: 11**
KiCad E.D.A. 10.0.6		Id: 1/8

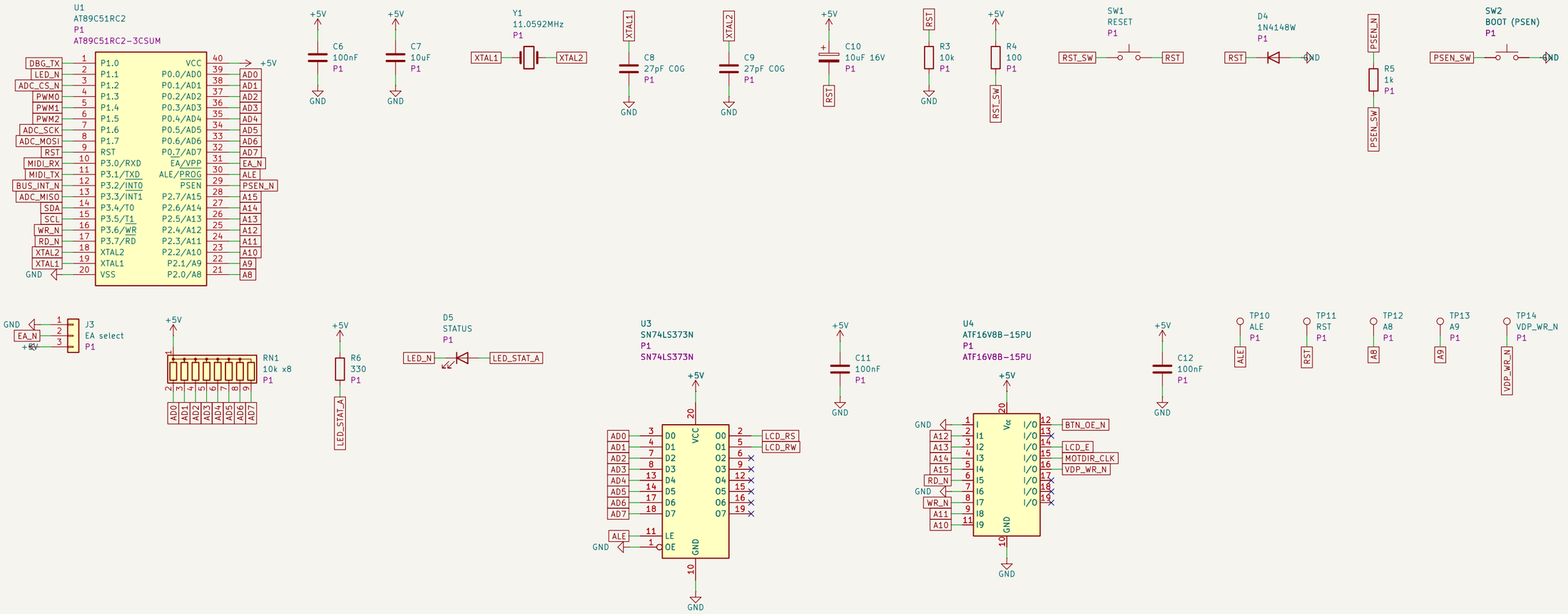


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Sheet: /Power/
File: power.kicad_sch

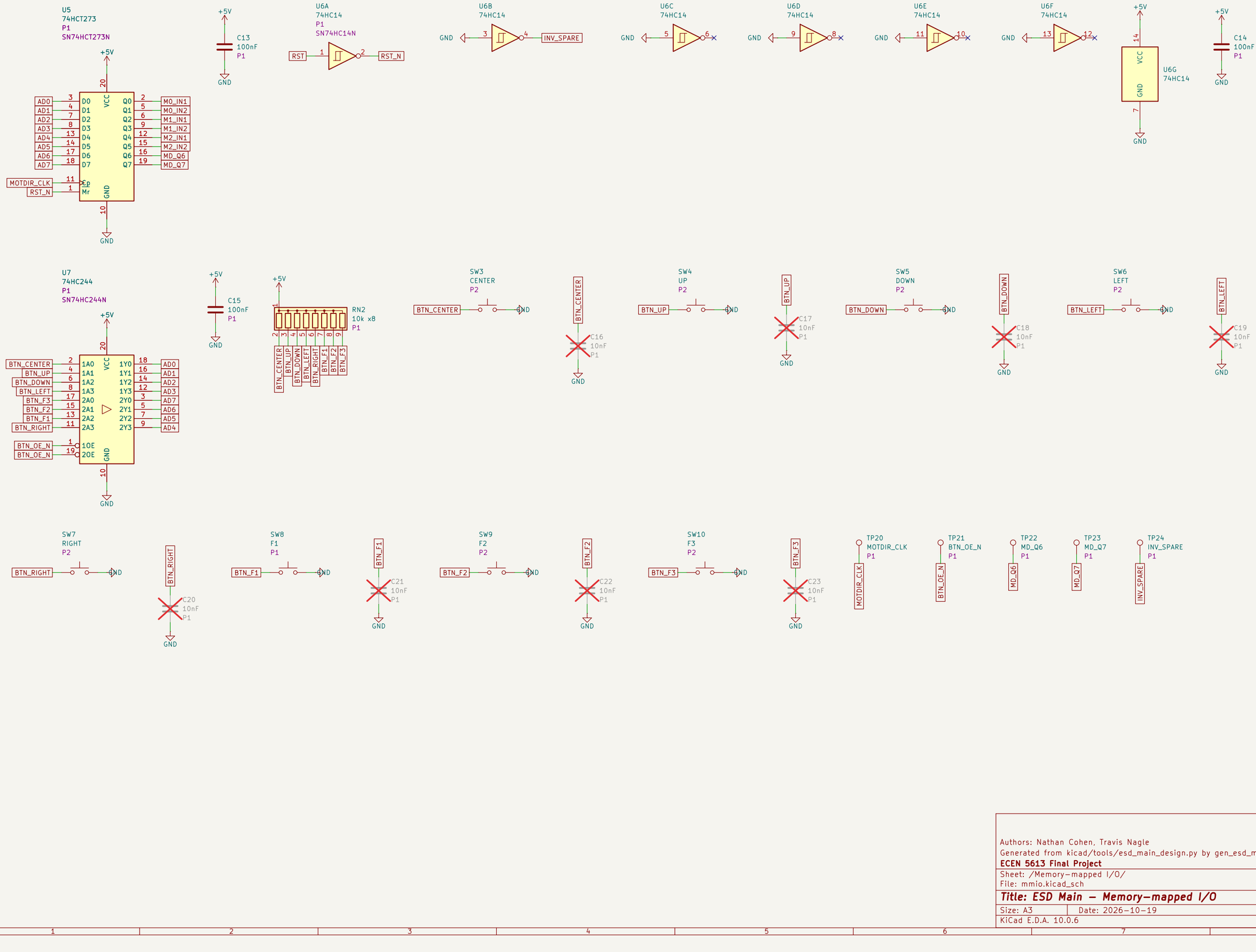
Title: ESD Main — Power

Size: A3	Date: 2026-10-19	**Rev: 11**
KiCad E.D.A. 10.0.6		Id: 2/8



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Sheet: /MCU + bus/		
File: mcu_bus.kicad_sch		
Title: ESD Main — MCU + bus		
Size: A3	Date: 2026-10-19	**Rev: 11**
KiCad E.D.A. 10.0.6		Id: 3/8

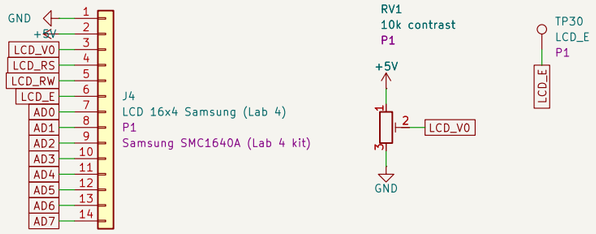


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Sheet: /Memory-mapped I/O/
File: mmio.kicad_sch

Title: ESD Main – Memory-mapped I/O

Size: A3	Date: 2026-10-19	**Rev: 11**
KiCad E.D.A. 10.0.6		Id: 4/8

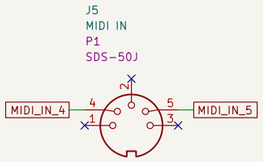


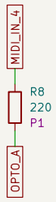
Authors: Nathan Cohen, Travis Nagle
Generated from kicad/tools/esd_main_design.py by gen_esd_main.py
ECEN 5613 Final Project

Sheet: /LCD/
File: lcd.kicad_sch

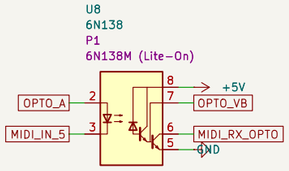
Title: ESD Main – LCD

Size: A3	Date: 2026-10-19	**Rev: 11**
KiCad E.D.A. 10.0.6		Id: 5/8





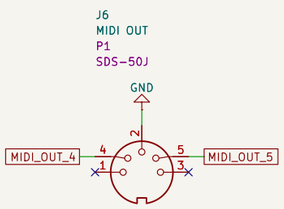


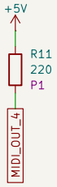




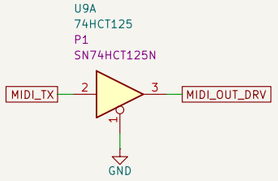


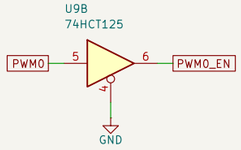


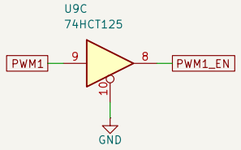


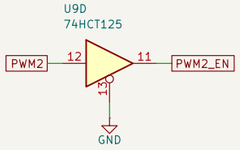


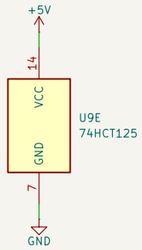




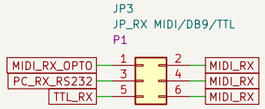


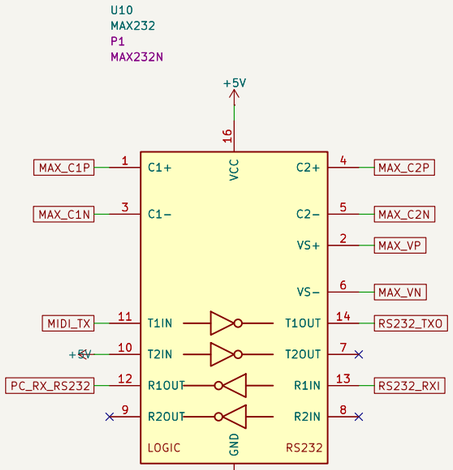


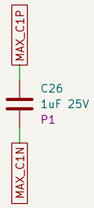


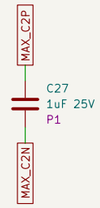


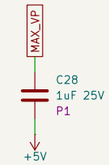


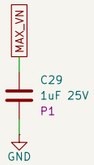




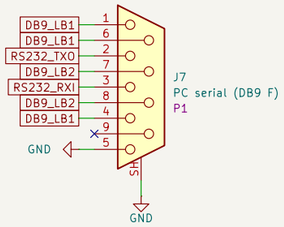


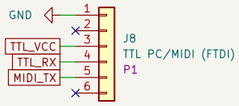




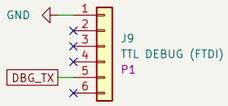


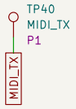












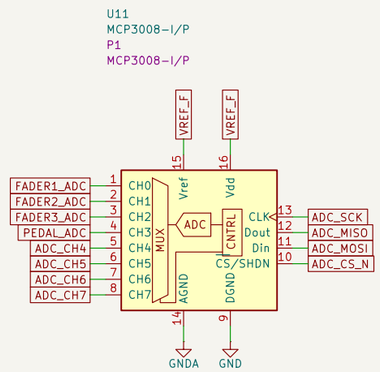


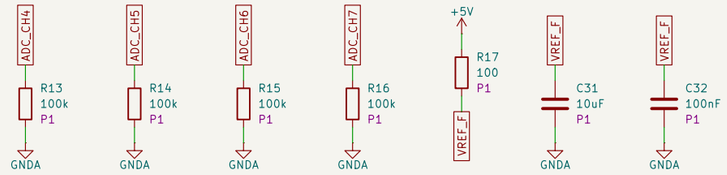
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Sheet: /MIDI + serial/
File: midi_serial.kicad_sch

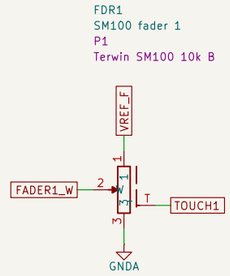
Title: ESD Main — MIDI + serial

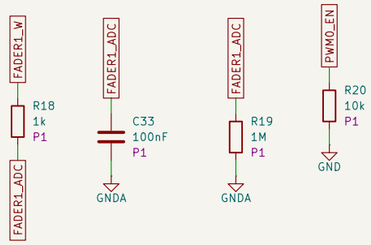
Size: A3	Date: 2026-10-19	**Rev: 11**
KiCad E.D.A. 10.0.6		Id: 6/8

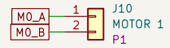


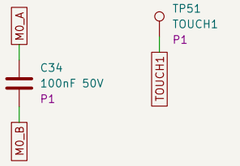


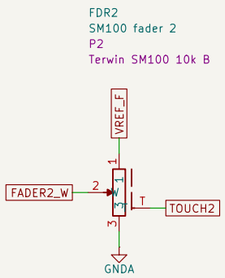


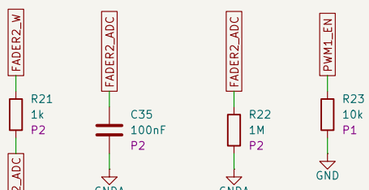


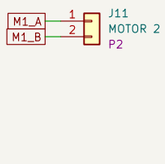


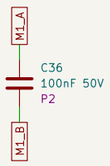


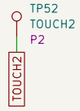


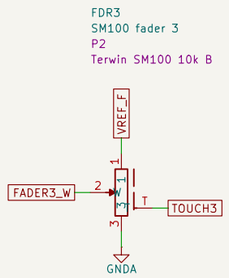


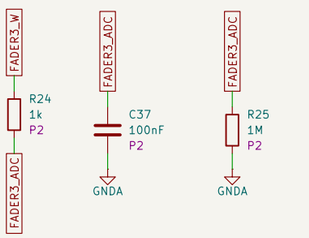


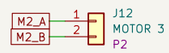


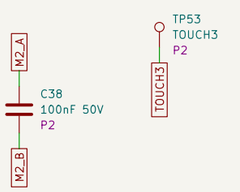


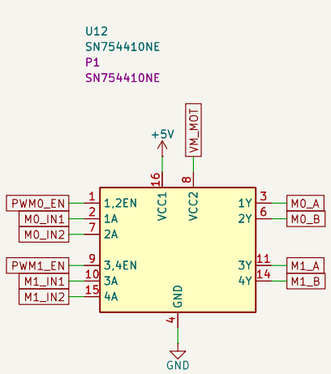


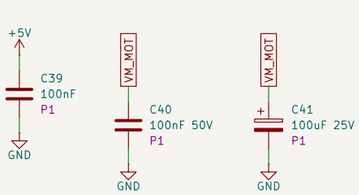


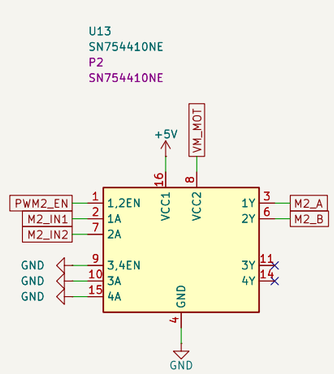


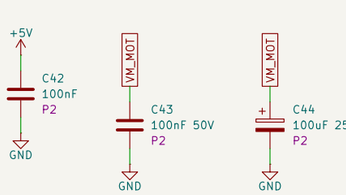


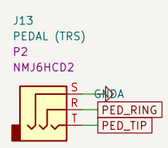


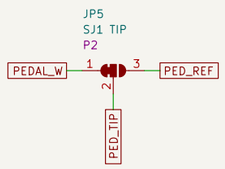


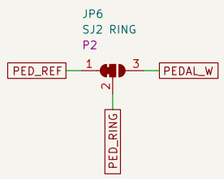




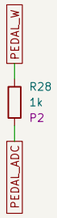


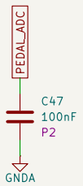


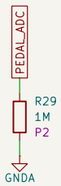












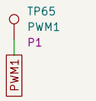
















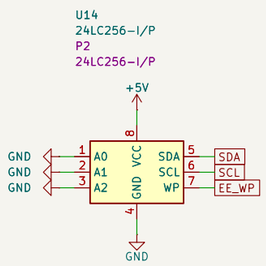


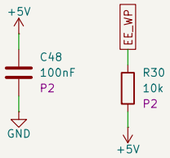
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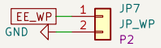
Sheet: /Faders + motor drivers/
File: faders.kicad_sch

Title: ESD Main — Faders + motor drivers

Size: A3	Date: 2026-10-19	**Rev: 11**
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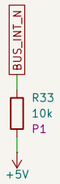


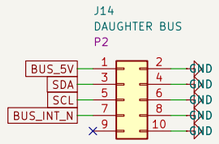


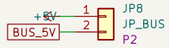


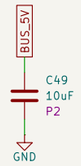


















Authors: Nathan Cohen, Travis Nagle
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ECEN 5613 Final Project

Sheet: /I2C + daughter bus/
File: i2c_bus.kicad_sch

Title: ESD Main — I2C + daughter bus

Size: A3	Date: 2026-10-19	**Rev: 11**
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